

Partitioning the Effect of Release and Liming on Growth of Sugar Maple and American Beech Saplings

Louis Duchesne, Jean-David Moore, and Rock Ouimet

ABSTRACT

In northeastern North America, a growing number of studies report on the regeneration failure of sugar maple (SM, *Acer saccharum* Marsh.) in some SM-dominated stands coupled with a marked increase in abundance of American beech (AB, *Fagus grandifolia* Ehrh.) in the regeneration stratum, suggesting change in forest composition toward AB dominance. The effect of release (removing all competitors within 1 m of their crown perimeters) and liming (3 Mg ha⁻¹ of CaCO₃) on growth of SM and AB saplings was experimentally tested to partition the effect of intertree competition and soil fertility on the growth dynamics of these two species at the sapling stage. Lime application had the desired effect on soil chemistry, expressed notably as a four-fold increase in calcium concentrations of the upper soil layer (0–12 cm) and a 63% decrease in exchangeable acidity. Seven years following treatments, biomass growth response of AB saplings to release was 1.5 times higher than that of SM. In contrast, liming increased biomass growth of SM saplings (approximately 2 times compared with unlimed saplings), while there was no effect on AB growth. Also, the availability of light and of soil Ca interacts to increase growth of SM saplings. Our results confirm that SM is more sensitive to calcium availability than AB. Various forest management strategies are discussed in light of these experimental results.

Keywords: sugarbush, decline, forest management, thinning, fertilization

Sugar maple (SM, *Acer saccharum* Marsh.) is one of the most abundant hardwoods in eastern Canada and the northeastern United States (Godman et al. 1990). Given that this shade-tolerant species is able to regenerate under its own canopy and forms uneven-aged structured forests, it has been historically categorized as an important late-successional species for large areas in southern Québec (Marie-Victorin 1964). Despite the absence of scientific consensus on causal factors, symptoms of SM decline and dieback of mature trees have been observed at different scales in the deciduous forest of northeastern North America since the late 1970s (Horsley et al. 2000, 2002, Duchesne et al. 2002). Additionally, a growing number of studies report the regeneration failure of SM in SM-dominated stands coupled with a marked increase in abundance of American beech (AB; *Fagus grandifolia* Ehrh.) in the regeneration stratum (Pothier 1996, Jenkins et al. 1999, Duchesne et al. 2005), suggesting future change in forest composition toward AB dominance. From an economic perspective, increased abundance of AB in these stands may be problematic because of its low value compared with SM. Moreover, SM has been historically exploited for maple syrup. Because the province of Québec is the largest producer of maple products, accounting for approximately 80% of global production (Government of Québec 2009), preserving the SM component is particularly important for maple syrup producers who

have made important infrastructure investments for sap collection and transformation. Increasing abundance of AB may also increase stand susceptibility to beech bark disease known as a massive ecosystem change driver in AB dominant forest (Griffin et al. 2003, Hane 2003).

From multiple-stand studies addressing this issue, it has been hypothesized that SM regeneration failure and increase in abundance of AB in some SM-dominated stands in Québec may result from one or a combination of factors, including (1) a better adaptation of AB to impoverished soils that are no longer adequate for SM recruitment, development, and competitiveness (Duchesne et al. 2005); (2) a partial canopy disturbance history that favored the regeneration of AB relative to SM (Nolet et al. 2008); and (3) a combination of multiple environmental changes that arose in SM-dominated forests over the past several decades (Gravel et al. 2011). Other factors such as the deep shade of old-growth stands (Beaudet et al. 1999), the shade cast by beech understories (Hane 2003), herbivore preference for SM in comparison to AB (Marquis and Brennemann 1981), and the allelopathic effects of beech foliage (Hane et al. 2003) have also been suggested as possible causes of SM regeneration failure and increased AB abundance.

Unfortunately, adequate forest management strategies to maintain SM representation in stands where a shift in recruitment from

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This article uses metric units; the applicable conversion factors are: centimeters (cm): 1 cm = 0.39 in.; meters (m): 1 m = 3.3 ft; square meters (m²): 1 m² = 10.8 ft²; millimeters (mm): 1 mm = 0.039 in.; kilometers (km): 1 km = 0.6 mi; hectares (ha): 1 ha = 2.47 ac; megagrams (Mg): 1 Mg = 2,204.6 lb.

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SM to AB has occurred cannot easily be drawn from these hypotheses. To assist juvenile growth of SM, foresters have to balance growth conditions to favor SM at the expense of AB. From the silvicultural toolbox, release of the sapling stratum may increase the available growing space for SM saplings, while liming may increase soil fertility. By modifying stem density and soil fertility, foresters can modify the composition of the sapling stratum and increase the growth, quality, and health of residual SM saplings.

The objectives of this study were to (1) partition the effect of intertree competition and soil fertility on growth of established SM and AB saplings in the understory of SM-dominated stands; and (2) propose silvicultural treatments aimed at promoting SM growth and development in the understory of SM-dominated stands where regeneration failure and an increase in the abundance of AB saplings relative to SM are observed.

Materials and Methods

Study Site

The study site is located at the Duchesnay Experimental Forest, approximately 50 km northwest of the city of Québec, QC. The elevation varies between 300 and 320 m. The average slope is approximately 10%. The mean annual temperature and annual precipitation are 3.4° C and 1,300 mm, respectively. The soil is classified as Orthic Ferro-Humic Podzol (Soil Classification Working Group 1998) belonging to the Sergeant soil series (Raymond et al. 1976). The humus is of moder type, and the parent material is a very acidic and stony glacial till derived from granitic gneiss bedrock. Soils in the area are typically 50–60 cm depth (above the C horizon) and are characterized by a sandy loam texture with approximately 45% of coarse fragment larger than 5 cm. More details on typical soil profile can be found in Ouimet and Duchesne (2005). Previous studies in the area demonstrated that (1) the soil calcium pool has been detrimentally affected by atmospheric acid deposition (Houle et al. 1997); (2) stem density, basal area, growth, and litter production of the SM population have been declining since the late 20th century (Duchesne et al. 2002, Duchesne et al. 2010); (3) the presence of AB has increased considerably in the regeneration stratum at the expense of SM (Duchesne et al. 2005); and (4) SM decline in terms of growth, nutrition, crown vigor, and regeneration may be counteracted by soil liming (Moore and Ouimet 2006, Moore et al. 2008). The forest on the experimental site is dominated by SM in association with yellow birch (YB, *Betula alleghaniensis* Britt.) and AB, with an estimated basal area of 21.6 m² ha⁻¹ for all stems having a diameter (1.3-m height from the highest root, DBH) larger than or equal to 11 mm (Table 1). The stand, which has been selectively exploited over the last century, was uneven aged and characterized by a large number of stems in the sapling stratum, which is mainly composed of AB and SM (see results section). The establishment of a dense quasi-even-aged AB cohort in the under-

story of an uneven-aged SM stand was partly attributable to the lower vigor and higher mortality rate of SM trees (Duchesne et al. 2005, Duchesne et al. 2010). The presence of such a cohort indicates there was relatively low competition occurring from declining overhead mature SM trees.

Experimental Setup and Sampling

The experimental setup consisted of a split-plot, 2 × 2 × 2 randomized design, replicated in two blocks arranged perpendicular to the slope gradient. Each block (approximately 25 m × 50 m) included two levels of release treatment as main plots (released versus unreleased), two levels of lime treatment (limed versus unlimed), and two species (SM and AB). In each block, we selected 20 SM and 20 AB saplings separated from each other by at least 5 m, for a total of 80 experimental units. In September 2004, 10 of the 20 saplings of each species in each block were randomly chosen and fertilized with 3 Mg ha⁻¹ of agricultural grade CaCO₃ applied manually over a 1.5-m radius around the trees. Five fertilized and five nonfertilized saplings of both species were then released from competition by removing all competitors within 1 m of their crown perimeters. Released trees were grouped within each block to limit possible side effects of the release treatment on unreleased trees. Sapling DBH were measured with an electronic caliper and the location of each measurement was identified with a marker to ensure the consistency of subsequent measurements, taken each year in October–November from 2005 to 2011. Sapling height was also measured yearly using a telescopic height-measuring pole.

Soils were sampled at two locations around each tree in both September 2004 before treatment application and again in 2009. The first soil surface samples (0–12 cm) were extracted using a 220-cm³ volumetric hammer soil sampler. Care was taken to avoid compacting the soil during sampling. This first sample contained the forest floor (7–10 cm in thickness), a small eluviated A horizon when present, and/or the top of the first B horizon. The mineral soil at 12–24 cm depth was qualitatively sampled separately using a bucket auger. Samples of the same depth were pooled for each tree, air-dried, weighed, and sieved to 2 mm before chemical analysis. Soil pH was measured using a soil-to-water ratio of 1:2.5. Exchangeable cations (K, Ca, Mg, and Al) were extracted with an unbuffered NH₄Cl (1 M, 12 h) solution and measured with an inductively coupled plasma emission spectrophotometer (Thermo Jarrell Ash ICAP-9000). Exchangeable hydrogen ions were directly measured in the extracts using a pH probe. Subsamples were ground to 0.5 mm for analysis of total carbon (dry combustion) and nitrogen (Kjeldahl; Page et al. 1982). Cation exchange capacity (CEC) was calculated as the sum of exchangeable cations and exchangeable hydrogen (units: cmol (+) kg⁻¹). Exchangeable acidity (H+Al) was calculated as the sum of exchangeable Al and exchangeable hydrogen. Base saturation (BS) was calculated as the percentage of CEC as base cations (Ca, Mg, K). Calcium saturation (CaS) was calculated as the percentage of CEC as calcium.

In each block, three circular plots (area = 400 m², radius = 11.28 m) were sampled in September 2004 for stand characterization. In each plot, every tree having a DBH larger than 90 mm was classified according to its species, and its DBH was recorded to the nearest mm. Trees with DBH from 11 to 90 mm were counted, classified, and measured in 40-m² circular subplots (radius = 3.57 m).

Table 1. Stand characteristics.

Species	Tree stratum (DBH > 90 mm)		Sapling stratum (11 mm ≤ DBH ≤ 90 mm)	
	Stems ha ⁻¹	m ² ha ⁻¹	Stems ha ⁻¹	m ² ha ⁻¹
Sugar maple	178	12.3	2,458	0.7
American beech	44	5.3	2,625	2.0
Yellow birch	5	1.0	125	0.1
Striped maple	0	0.0	83	0.1
Total	227	18.7	5,292	2.9

Table 2. Effect of lime and soil depth on soil properties in 2009 (adjusted means and standard error between parentheses), and F values of selected treatment effects.

Soil depth	Lime	pH	C org ¹	K	Ca	Mg	H + Al	CEC	BS ²	CaS ³	Ca/Mg	K/Mg
(cm)	(Mg ha ⁻¹)		(g kg ⁻¹)			(cmol (+) kg ⁻¹)			(%)			
0–12 ⁴	0	4.49 (0.04)	124.3 (7.3)	0.44 (0.02)	2.33 (0.25)	0.36 (0.03)	5.4 (0.4)	7.8 (0.7)	37.5 (1.5)	26.4 (1.2)	5.4 (0.5)	1.21 (0.04)
	3	4.73 (0.04)	139.4 (6.8)	0.44 (0.02)	9.17 (0.69)	0.45 (0.03)	2.0 (0.3)	12.1 (0.6)	77.7 (2.5)	69.3 (2.7)	17.8 (1.2)	1.06 (0.04)
12–24	0	4.69 (0.04)	92.2 (5.3)	0.22 (0.02)	0.40 (0.06)	0.19 (0.02)	3.7 (0.2)	4.6 (0.4)	18.3 (1.0)	9.8 (0.7)	3.2 (0.4)	1.09 (0.04)
	3	4.71 (0.03)	89.9 (5.1)	0.21 (0.02)	0.86 (0.10)	0.19 (0.02)	3.5 (0.2)	4.9 (0.3)	26.4 (1.4)	18.9 (1.2)	7.0 (0.9)	1.13 (0.04)
	ndf/ddf ⁵						F (P) values					
Soil depth	1/127	7 (0.008)	60 (<0.001)	78 (<0.001)	122 (<0.001)	67 (<0.001)	3 (0.076)	75 (<0.001)	389 (<0.001)	379 (<0.001)	92 (<0.001)	0.1 (0.763)
Liming	1/127	10 (0.002)	0.1 (0.718)	0.2 (0.640)	41 (<0.001)	0.3 (0.612)	19 (<0.001)	6 (0.016)	146 (<0.001)	161 (<0.001)	110 (<0.001)	6 (0.018)
Soil depth × liming	1/127	13 (<0.001)	4 (0.052)	0.6 (0.433)	76 (<0.001)	9 (0.003)	37 (<0.001)	20 (<0.001)	117 (<0.001)	120 (<0.001)	52 (<0.001)	13 (<0.001)

Note: Although the tree species and release treatments were included in the statistical analyses, they are not reported in this ANOVA table.

¹C org: organic carbon; ²BS: base saturation; ³CaS: calcium saturation; ⁴contains the forest floor + top mineral layer; ⁵ndf: numerator degrees of freedom; ddf: denominator degrees of freedom.

Data Handling and Statistical Analyses

Stand basal area (m² ha⁻¹) and stem density (stem ha⁻¹) were computed from the six circular sampling plots. Sapling wood biomass was estimated using the Canadian national DBH- and height-based species-specific allometric aboveground biomass equations developed by Lambert et al. (2005). Basal area, height, and biomass annual measurements of each sapling were expressed as percentage of their initial value measured in 2004. The effect of time, species, lime and release treatments, and their interactions on relative stem basal area, height, and biomass were examined by analysis of variance (ANOVA) using the PROC GLIMMIX procedure with a first-order autoregressive variance structure for random effects and correlated errors. Polynomial contrasts were built to assess the effect of time, species, treatments, and interactions. Soil data were analyzed using a generalized least squares (gls) model of ANOVA including soil depth, species, lime and release treatments, and their interactions as independent factors, allowing for heterogeneity correction with differing variances per stratum and with the respective 2004 values as covariates. The “nlme” package of the R software (Pinheiro et al. 2010) was used to perform these analyses.

Results

Initial Stand Characteristics

The studied stand was dominated by SM, AB, and YB, representing 60, 34, and 5%, respectively, of the total basal area (21.6 m² ha⁻¹; Table 1). The stand contained a large density of saplings (5,292 stem ha⁻¹) of which SM and AB represented respectively 46 and 50%. While SM represented 78% of the total number of stems and 66% of the total basal area of the tree stratum, its proportion decreased to 46 and 24% in terms of stem density and basal area, respectively, in the sapling stratum (Table 1). For a roughly equivalent stem density, total basal area of AB saplings was approximately 2.9 times higher than for SM saplings, confirming that SM saplings were much smaller than AB in terms of DBH.

Soil

Five years after the lime treatments, many soil properties had changed (Table 2). Among the most important changes that occurred within the first 12 cm of soil was the average concentration of exchangeable Ca of limed soils reaching approximately 4 times those of the controls. Consequently, CEC, BS, CaS, and the Ca/Mg ratio were 1.5–3.3 times higher, while H+Al concentrations were 63% lower in limed soils than in the controls. The pH of the 0–12 cm soil layer was slightly higher in the limed treatment ($P < 0.001$),

while the K/Mg ratio slightly lower ($P < 0.001$). Organic C concentrations were slightly higher in the limed soil 0–12 cm layer compared to the controls ($P = 0.05$). However, neither N concentrations (mean $\pm$ se: 8.71 ± 0.36 g kg⁻¹ in the 0–12 cm layer; 4.35 ± 0.18 g kg⁻¹ in the 12–24 cm layer), bulk density of the 0–12 cm layer (0.293 ± 0.008 g cm⁻³), nor the C/N ratio (17.36 ± 0.14 in both the 0–12 and 12–24 cm layers) were influenced by the lime treatment after 5 years ($P \geq 0.06$).

The effect of lime was less important in the deeper soil layer (12–24 cm). Concentrations of exchangeable Ca in limed soils at this depth were approximately twice the concentrations of the controls (Table 2). In this soil layer, BS, CaS, and the Ca/Mg ratio were respectively 1.4, 1.9, and 3.2 times higher in limed soils than in the controls, while no change was observed for CEC, exchangeable acidity, pH, K/Mg ratio, and organic C and N concentrations 5 years after liming ($P \geq 0.09$).

Some soil parameters (pH, C/N ratio, and H+Al concentrations and Ca/Mg ratio) showed statistical effects associated with tree species or the release treatment ($P \leq 0.05$), but the differences were relatively small and probably of no ecological significance.

Sapling Growth

The sampled saplings reflected the dominance of AB in terms of sapling size. In 2004, stem DBH and height of AB saplings were 31 ± 1.0 mm and 5.3 ± 0.11 m, respectively, while DBH and height of SM saplings were estimated to be 22 ± 0.9 mm and 4.3 ± 0.14 m, respectively. Basal area growth response to release differed between species ($P < 0.001$, for species $\times$ release $\times$ years interaction, Figure 1, Table 3). Seven years after treatment, the basal area of released SM and AB saplings had increased by 73 and 217%, respectively, while the basal area of unreleased saplings had increased by 38 and 94%, compared to their respective basal areas before treatment. For both species, relative growth of released saplings after 7 years was approximately 2 times higher (1.9 times for SM and 2.3 times for AB) than for unreleased saplings. Basal area growth response to liming also differed between species ($P = 0.004$ for species $\times$ liming $\times$ years interaction, Figure 1, Table 3). In contrast to the growth response to release, only SM saplings responded to liming. After 7 years, basal area of limed SM saplings increased by 75%, while unlimed saplings increased by 37%, compared to their basal area before treatment. Growth of limed and unlimed AB saplings did not differ significantly ($P = 0.73$), increasing by 156 and 155%, respectively, compared to their basal area before treatment.

Height-growth response to release differed between species ($P < 0.001$, for species $\times$ release $\times$ year interaction, Figure 2, Table 3),

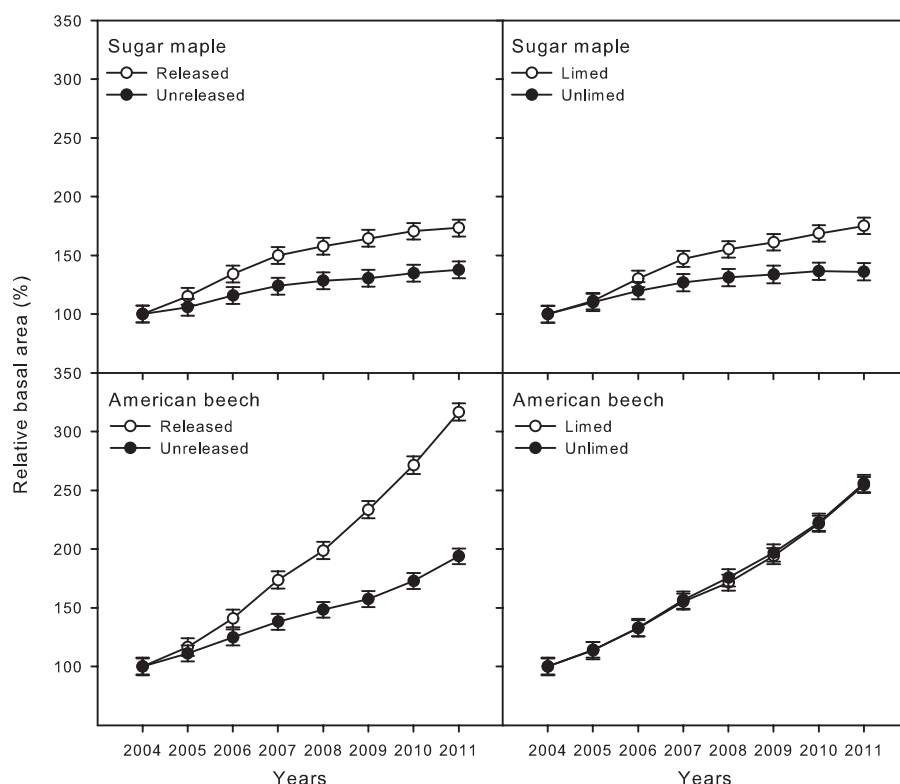


Figure 1. Relative basal area growth of SM and AB saplings to release and liming. Error bars represent $\pm$ standard error.

Table 3. Summary of ANOVA results for basal area, height, and biomass growth response of SM and AB saplings to liming and release.

Source of variation	ndf/ddf	Relative basal area		Relative height		Relative biomass	
		F	P	F	P	F	P
Species (S)	1/61	43.4	< 0.001	21.0	< 0.001	20.5	< 0.001
Release (R)	1/61	49.6	< 0.001	0.3	0.562	31.5	< 0.001
Liming (L)	1/61	2.9	0.093	0.4	0.542	3.1	0.084
Year (Y)	7/427	283.0	< 0.001	120.3	< 0.001	248.4	< 0.001
S $\times$ R	1/61	6.6	0.013	4.8	0.032	8.0	0.006
S $\times$ L	1/61	3.8	0.056	0.2	0.685	3.9	0.052
S $\times$ Y	7/427	66.3	< 0.001	30.6	< 0.001	40.4	< 0.001
R $\times$ L	1/61	0.3	0.589	0.4	0.510	0.3	0.611
R $\times$ Y	7/427	43.0	< 0.001	1.7	0.119	28.8	< 0.001
L $\times$ Y	7/427	2.4	0.021	0.8	0.587	2.9	0.006
S $\times$ R $\times$ Y	7/427	15.1	< 0.001	4.0	< 0.001	14.3	< 0.001
S $\times$ L $\times$ Y	7/427	3.1	0.004	0.6	0.795	3.5	0.001
S $\times$ R $\times$ L	7/427	0.8	0.387	0.3	0.622	0.5	0.497
R $\times$ L $\times$ Y	7/427	0.2	0.994	0.3	0.954	0.2	0.985
S $\times$ R $\times$ L $\times$ Y	7/427	1.1	0.384	0.4	0.890	0.6	0.744
Contrasts (S $\times$ R $\times$ Y)			P		P		P
SM released SM unreleased			< 0.001		< 0.001		0.002
AB released AB unreleased			< 0.001		0.004		< 0.001
Contrasts (S $\times$ L $\times$ Y)							
SM limed versus SM unlimed			< 0.001		0.013		< 0.001
AB limed versus AB unlimed			0.726		0.656		0.743
Contrasts (S $\times$ R $\times$ L $\times$ Y)							
SM released versus SM controls			< 0.001		0.751		0.002
SM limed versus SM controls			< 0.001		0.029		< 0.001
SM released and limed versus SM controls			0.001		< 0.001		< 0.001
AB released versus AB controls			< 0.001		0.315		< 0.001
AB limed versus AB controls			0.132		0.459		0.205
AB released and limed versus AB controls			< 0.001		< 0.001		< 0.001

Note: Bold indicate significance ($P < 0.05$); ndf: numerator degrees of freedom; ddf: denominator degrees of freedom; SM: sugar maple; AB: American beech.

while there was no effect of the liming treatment on overall height growth ($P = 0.79$, for species $\times$ liming $\times$ years interaction). Over the 7-year period, height growth of released SM (8%) was lower than height growth of unreleased SM saplings (15%; $P < 0.001$). In

contrast, released AB saplings grew faster (41%) in height than unreleased AB saplings (32%; $P = 0.004$).

Biomass growth response to release and liming also differed between species ($P \leq 0.001$, for species $\times$ release $\times$ years interaction

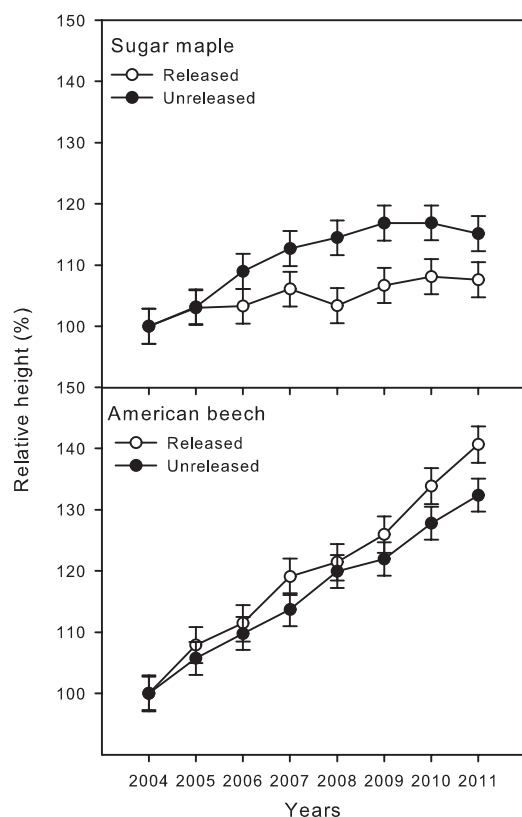


Figure 2. Relative height growth of SM and AB saplings to release. Error bars represent $\pm$ standard error.

and for species $\times$ liming $\times$ years interaction, Figure 3, Table 3). Seven years following treatment, released SM and AB saplings increased in biomass by 86 and 231%, respectively, while the biomass of unreleased saplings increased by 59 and 100%, respectively, compared to biomass before treatment. Relative biomass growth of released SM and AB saplings after the 7-year period was respectively 1.5 and 2.3 times higher than for unreleased saplings (Figure 3, Table 3). As for basal area growth, only SM saplings exhibited increased biomass in response to liming. After 7 years, biomass of limed SM saplings increased by 99% while for unlimed saplings it increased by 47%, compared to biomass before treatment. Biomass growth of limed and unlimed AB saplings did not differ significantly ($P = 0.74$), increasing by 167 and 165%, respectively.

Relative biomass growth responses to liming, release, and the combination of both treatments are shown in Figure 4. Despite the absence of a significant species $\times$ liming $\times$ release $\times$ years interaction, specific contrasts indicated that over the period, enhanced biomass growth of SM saplings relative to controls was 23% for release ($P = 0.002$), 47% for liming ($P < 0.001$), and 79% when both treatments were combined ($P < 0.001$; Table 3). Relative to controls, the effect of release was perceptible the first year following treatment, but SM biomass growth enhancement was of short duration, the differences being more or less stable after 2 years. In contrast, SM biomass growth enhancement by liming was observable the second year following treatment, and its effect increased linearly throughout the study period. For AB, biomass growth enhancement relative to controls was negligible (14%) for liming ($P = 0.21$), 147% for release ($P < 0.001$), and 130% when both treatments were combined ($P < 0.001$). Biomass growth of limed AB saplings did not differ from controls, while their growth response to

combined treatments did not differ significantly from that due to cleaning only ($P = 0.88$).

Discussion

Soil Response to Liming

Lime application had the desired effect on soil chemistry without undue alteration of nontargeted soil properties. We observed a four-fold increase in calcium concentration of the upper soil layer (0–12 cm) and a 63% decrease in exchangeable acidity. Similar results have been reported for a nearby site 5 years after a dolomitic lime application at a rate of 5 Mg ha^{-1} (Houle et al. 2002).

The lime application rate used (3 Mg ha^{-1}) did not strongly influence soil pH. According to a previous study in this area (Moore et al. 2008), more than 10 Mg ha^{-1} of dolomitic lime (or more than 13 Mg ha^{-1} of CaCO_3 lime) would have been needed to increase soil pH by one unit in the first 12 cm (including the forest floor) of this forest soil. This result suggests that soil pH does not appear to be a good indicator of soil fertility response following lime addition. Base cation availability, particularly Ca in this study, appears to be a better indicator.

The lime treatment slightly increased organic C concentrations in the topsoil layer as already observed in a similar and close SM stand (Houle et al. 2002). This accumulation of organic C may be explained in part by the saturation of the forest humus in Ca, causing the flocculation of humic acids, their stabilization, and increasing their resistance to biodegradation (Duchaufour 1990). A more favorable soil chemical environment may also have promoted fine root growth, which is a major source of organic matter in forest soils. However, the long-term effect of liming on soil organic matter status still remains a matter of speculation (Haynes and Naidu 1998).

Application of calcareous lime in this experiment slightly lowered the K/Mg ratio in the upper soil layer but by a smaller magnitude than in other experiments using dolomitic lime (Long and Horsley 2011: 22 Mg ha^{-1} , Moore et al. 2000: up to 50 Mg ha^{-1}). The soil K/Mg ratio is indicative of SM foliar K status (Ouimet and Camiré 1995). In the two above-mentioned studies, K-Mg antagonism from liming with dolomitic lime was noted, resulting in the lowering of foliar K concentrations down to or below some critical thresholds (cf. Moore and Ouimet 2010). In our study, it appears that calcareous lime did not cause a strong imbalance between available K and Mg in the soil. Therefore, the K status of SM saplings might not have been altered by calcareous lime application.

Sapling Growth Response to Liming

Our results revealed that SM responded to liming by doubling its biomass increment (99%) throughout the study period compared to unlimed SM saplings (47%), while AB did not show any growth response to this treatment (Figure 3). Sugar maple is known to be sensitive to the base status of soils. Beneficial effects of Ca addition on SM regeneration were noted in many studies in northeastern North America (e.g., Kobe et al. 2002, Juice et al. 2006, Bigelow and Canham 2007). Higher-base cation concentrations of soil, foliage, or stem tissues have been linked to lower SM crown defoliation rates (e.g., Côté and Camiré 1995, Houle et al. 2007, Hallett et al. 2006). Tree-growth decline, as deduced from tree core measurements, has also been related to the base status of soils (e.g., Heisey 1995, Wilmot et al. 1995, Long et al. 2009). The high sensitivity of SM to the base status of soils has also been documented experimentally in both field and controlled environments. While decline in

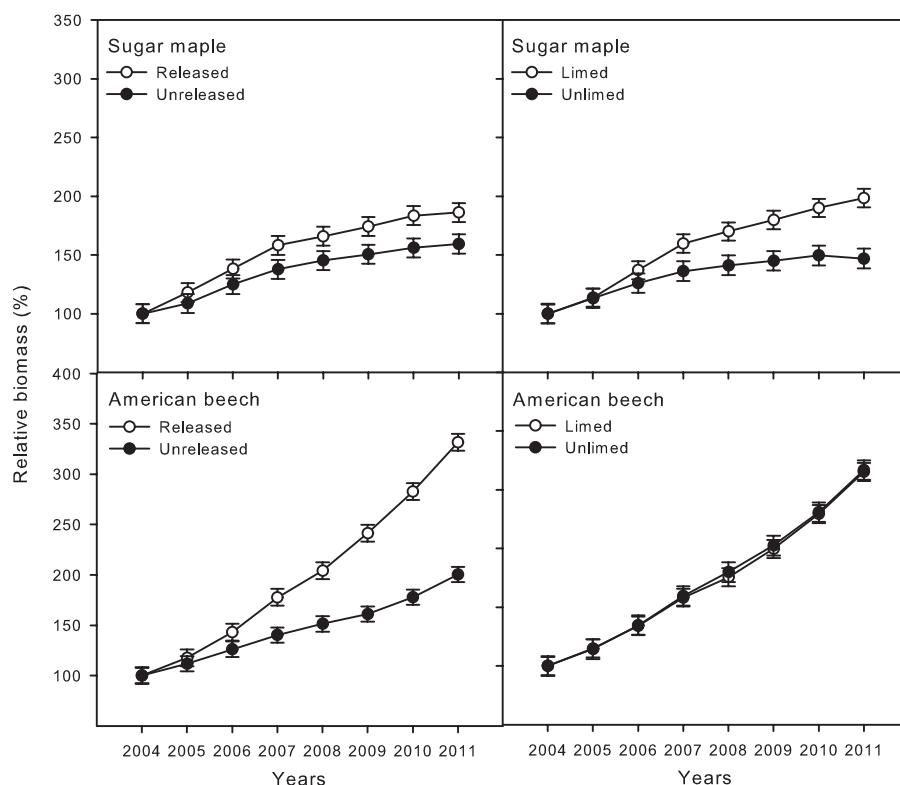


Figure 3. Relative biomass growth of SM and AB saplings to release and liming. Error bars represent $\pm$ standard error.

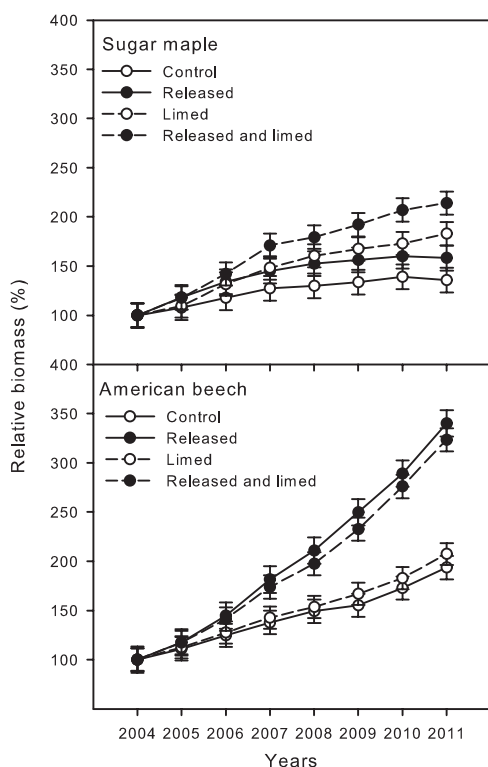


Figure 4. Relative biomass growth of control SM and AB saplings and their response to release, liming, and both treatments combined. Error bars represent $\pm$ standard error.

growth and vigor of SM may be induced by soil acidification (Ouimet et al. 2008, Bethers et al. 2009), alkalization of acidic soils has shown that SM health and growth are limited by low soil levels of

Ca (and Mg; e.g., Juice et al. 2006, Moore and Ouimet 2006, Long and Horsley 2011).

Our results are in accordance with a growing number of studies reporting that SM is generally more sensitive to Ca availability than AB. Unlike SM, AB does not show any response to calcium addition (Long et al. 1997, Kobe et al. 2002, Bigelow and Canham 2007). In natural environments, it has been demonstrated that SM is generally more sensitive to nutrient availability than AB (Park and Yanai 2009). Survival of SM seedlings declined while that of AB increased at higher levels of soil exchangeable Al, and this pattern is consistent with the position of these species along a soil acidity gradient (Bigelow and Canham 2010). It has also been shown that the relative abundance of SM is positively related to soil Ca availability, while a negative relationship has been observed for AB (Page and Mitchell 2008, Duchesne and Ouimet 2009). Similarly, Schwarz et al. (2003) observed that the mean abundance of SM increased with pH and was negatively associated with increasing levels of exchangeable Al. Arian et al. (2005) concluded that low Ca availability was associated with greater dominance of AB. Some previous studies observed no significant effect of Ca addition on SM (e.g., Burke and Raynal 1998, Ouimet et al. 2008, Gasser et al. 2010), but this can be explained by (1) an insufficient addition of Ca; (2) a non-Ca-limited site; or (3) a combination of these two factors.

Sapling Growth Response to Release

In the exploration of possible mechanisms ensuring species codominance in maple-beech forests, some previous studies documented the growth response of these two species to light availability at the seedling and sapling stages. Some studies documented that growth of AB was higher than that of SM under closed canopy (Canham 1988, Poulson and Platt 1996), and that SM responded more markedly than AB to the increase in light availability (Canham

1988, Poulson and Platt 1996, 1989). Others documented that growth of AB seedlings was greater than of SM in both canopy-gap and closed-canopy conditions and that growth response to an increase in light availability was higher for AB than for SM (Takahashi and Lechowics 2008, Finzi and Canham 2000). Others also documented that AB grew faster than SM in canopy gaps and under a closed canopy but that sensitivity to light availability did not differ between the two species (Beaudet and Messier 1998, Beaudet et al. 2007). These divergent responses suggest that growth response of SM and AB saplings to light availability is largely variable among sites (Kobe et al. 1995, Beaudet and Messier 1998, Nolet et al. 2008).

In comparing growth-mortality functions of juvenile SM and AB for sites differing in terms of climate and soil characteristics, Kobe et al. (1995) concluded that SM response to light availability was relatively insensitive to climate but presumably more sensitive to soil nutrient availability. These authors have shown that the probability of SM mortality under low light decreased considerably on calcareous sites compared to other more acidic sites. Comparatively, they noted that AB appears to be less labile regarding shade tolerance than SM. Similarly, Zaccherio and Finzi (2007) observed that light interacted with calcium availability on growth of SM seedlings. On base-poor soil, diameter growth rate in response to Ca addition was much greater in canopy gaps than in the understory. Similar observations on greater growth response of Ca-fertilized SM trees to release from competition have been reported at the Hubbard Brook Experimental Forest in New Hampshire (Hugget et al. 2007). Our results also confirmed the interaction of light availability and soil Ca availability on growth of SM saplings. After 7 years, biomass of released-only SM saplings increased by 58%, limed-only SM saplings by 82%, and released and limed SM saplings by 114% (Figure 4). Consequently, growth response to release was greater for SM saplings growing on a richer soil. It may also be interpreted that SM growth response to an increase in Ca availability is greater for saplings subjected to less competition. In contrast, there was no change in the growth response of AB saplings to release, regardless of soil nutrient availability (Figure 4).

Interestingly, while AB had a positive height-growth response to release, height growth of released SM saplings was nearly half as great as that of unreleased SM saplings. In Pennsylvania, St. Clair and Lynch (2005) suggested that the regeneration and health of SM seedlings growing on acid and nutrient-poor soils may be negatively affected in canopy gaps where light may exacerbate nutrient stress. In our study, released SM saplings were initially suppressed and outcompeted by AB saplings. For a given height, trees with a competitive advantage should have a greater diameter than the trees that they have outcompeted. Therefore, SM saplings had a higher initial height/diameter ratio (203 ± 5) than AB saplings (174 ± 4). Crown bending, crown damage, and optimum species height/diameter ratio may all be involved in the negative height growth response of SM saplings to release. Indeed, height/diameter ratios are indicative of a tree's ability to withstand snow and ice loading as well as wind (Wonn and O'Hara 2001). Many released SM saplings, for which the crown was no longer laterally supported by surrounding sapling stems, tended to bend following release. On many occasions, weakened apical branches of released SM saplings may have also been broken by snow and ice loading. Released SM saplings should, therefore, grow more rapidly in terms of diameter than in terms of height to provide a stronger support for future crown development.

The higher growth response of AB to release as compared to SM was in line with previous studies addressing the effects of cleaning and early crop-tree release in northern hardwood stands (see review by Heitzman and Nyland 1991). It is generally recognized that healthy and dominant trees respond better to release than overtopped and low vigor ones.

Implications for Forest Management and Conclusion

In this study, we reported on the effect of release and liming on growth of SM and AB saplings in a SM-dominated stand located on base-poor acidic soil, with a dense cohort of AB recruits in the understory and poor SM regeneration. Over a 7-year period, biomass growth of control AB saplings (93%) was approximately 2.6 greater than for SM saplings (36%), indicating that, without silvicultural treatment in this ecosystem, SM is rapidly supplanted by AB saplings making up the future forest stand. Based on the pursued objectives, various forest management strategies may be considered in light of the experimental results. In terms of wood volume and biomass production, the release of evenly distributed AB saplings would be the most beneficial treatment to promote growth of the sapling stratum. This treatment will, however, accelerate changes in forest composition already perceptible in this SM-dominated stand. Releasing saplings of both species, without lime application, will increase the competitive advantage of AB saplings because the response of AB to release was greater than the response of SM. Releasing evenly distributed SM saplings will have a short-term effect and will result in a cohort mainly composed of weakened (height/DBH ratio = 203) and low-growth (58% in 7 years) SM saplings. Liming treatment was more favorable to SM growth (83%) than to AB growth, the latter species being nonresponsive. The effect of liming was, however, not sufficient to ensure that SM will be competitive against AB in this ecosystem. The combination of liming and release of SM saplings was the only treatment that provided a competitive advantage to SM (114%) over unreleased AB saplings (93%). Liming will also promote crown recovery and growth of declining mature SM trees, increased light interception, and promote SM seedling establishment (Moore et al. 2008). When the abundance and distribution of SM saplings are adequate, this treatment could be coupled with a strong reduction of the AB component in the sapling stratum. This treatment could be efficient in preserving the SM component, but it will inevitably have considerable detrimental effects on wood production and yield compared to the release of AB saplings. However, for maple syrup producers, this is currently the only proven way to preserve SM representativeness in acidic and base-poor sugarbushes. Given that these management strategies are based on the results of a relatively small experiment at only one site, we nevertheless suggest some caution in their application at an operational scale. They could, however, be included in an adaptive management approach where practitioners apply such treatments in a quasi-research manner, with elevated monitoring and reflection on whether the treatments really performed on their sites.

This experiment was carried out in a SM-dominated stand in which a dense cohort of AB was supplanting the SM regeneration. We can expect that applying the two treatments would be even more beneficial for SM if they were applied earlier in the sapling development stage, before AB takes advantage and supplants the SM saplings. Stronger SM saplings would probably react better to the treatments. To ensure the best growth response to release, candidate trees should represent the best individuals among the competing trees

and release should occur before they decline strongly in vigor (Heitzman and Nyland 1991).

It has been recently demonstrated that beneficial effects of fertilization with base cations on SM are still observable 15–23 years following lime application (Moore et al. 2012, Long et al. 2011). Given that added Ca is moving relatively slowly from the forest floor to the underlying mineral soil (Moore et al. 2012), we can expect long-term effects of a single lime application. On the other end, although release treatments may results in substantial short-term changes in density, structure, and species composition, it has been suggested that these effects will essentially disappear over few decades (Leak and Smith 1997). Nevertheless, long-term effects of the combination of both treatments on SM saplings remain to be documented. To address this question, measurements of this experiment will be pursued.

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